

Experiment 5 - Write a Python program to solve the Missionaries and Cannibals Problem.

Program Code

```
from collections import deque

def is_valid(m_left, c_left, m_right, c_right):
    if m_left < 0 or c_left < 0 or m_right < 0 or c_right < 0:
        return False

    if m_left > 0 and m_left < c_left:
        return False

    if m_right > 0 and m_right < c_right:
        return False

    return True

def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)

    queue = deque()
    visited = set()

    queue.append((start, []))

    moves = [
```

```
(1, 0),  
(2, 0),  
(0, 1),  
(0, 2),  
(1, 1)  
]
```

```
while queue:  
    state, path = queue.popleft()  
  
    if state in visited:  
        continue  
  
    visited.add(state)  
    path = path + [state]  
  
    m_left, c_left, boat = state  
    m_right = 3 - m_left  
    c_right = 3 - c_left  
  
    if state == goal:  
        return path  
  
    for m, c in moves:  
        if boat == 0:  
            new_m_left = m_left - m  
            new_c_left = c_left - c  
            new_boat = 1  
        else:  
            new_m_left = m_left + m
```

```
new_c_left = c_left + c
new_boat = 0
```

```
new_m_right = 3 - new_m_left
new_c_right = 3 - new_c_left
```

```
if is_valid(new_m_left, new_c_left, new_m_right,
new_c_right):
    new_state = (new_m_left, new_c_left, new_boat)
```

```
if new_state not in visited:
    queue.append((new_state, path))
```

```
return None
```

```
solution = solve()
```

```
if solution:
    print("Solution found!")
    print()
```

```
for step, state in enumerate(solution):
    m_left, c_left, boat = state
    m_right = 3 - m_left
    c_right = 3 - c_left
```

```
side = "Left" if boat == 0 else "Right"
```

```
print(
```

```

        "Step", step,
        ": Left =", m_left, "Missionaries,", c_left, "Cannibals |",
        "Right =", m_right, "Missionaries,", c_right, "Cannibals |",
        "Boat =", side
    )
else:
    print("No solution found.")

```

Program Output

```

exe "d:/AI LAB/experiment5_missionaries_cannibals.py"

```

```

Solution found!

```

```

Step 0 : Left = 3 Missionaries, 3 Cannibals | Right = 0 Missionaries, 0 Cannibals | Boat = Left
Step 1 : Left = 3 Missionaries, 1 Cannibals | Right = 0 Missionaries, 2 Cannibals | Boat = Right
Step 2 : Left = 3 Missionaries, 2 Cannibals | Right = 0 Missionaries, 1 Cannibals | Boat = Left
Step 3 : Left = 3 Missionaries, 0 Cannibals | Right = 0 Missionaries, 3 Cannibals | Boat = Right
Step 4 : Left = 3 Missionaries, 1 Cannibals | Right = 0 Missionaries, 2 Cannibals | Boat = Left
Step 5 : Left = 1 Missionaries, 1 Cannibals | Right = 2 Missionaries, 2 Cannibals | Boat = Right
Step 6 : Left = 2 Missionaries, 2 Cannibals | Right = 1 Missionaries, 1 Cannibals | Boat = Left
Step 7 : Left = 0 Missionaries, 2 Cannibals | Right = 3 Missionaries, 1 Cannibals | Boat = Right
Step 8 : Left = 0 Missionaries, 3 Cannibals | Right = 3 Missionaries, 0 Cannibals | Boat = Left
Step 9 : Left = 0 Missionaries, 1 Cannibals | Right = 3 Missionaries, 2 Cannibals | Boat = Right
Step 10 : Left = 1 Missionaries, 1 Cannibals | Right = 2 Missionaries, 2 Cannibals | Boat = Left
Step 11 : Left = 0 Missionaries, 0 Cannibals | Right = 3 Missionaries, 3 Cannibals | Boat = Right

```